

Corrected Inference for the CPE Portfolio Tilt Strategy: Newey-West HAC Standard Errors and Robustness Checks

Arun Ramanathan, Ph.D.
Independent Quantitative Research
quantarram.github.io | github.com/quantarram/quant-regime-research
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ABSTRACT

Paper 4 of the CPE series reported statistically significant portfolio tilt alpha at 63-day ($t=4.07$, $p<0.001$) and 126-day ($t=5.51$, $p<0.000001$) horizons using a standard one-sample t-test, but flagged overlapping weekly observations as a key limitation requiring Newey-West HAC-corrected inference. This research letter closes that gap. Applying HAC-robust standard errors with both rule-of-thumb (3 lags) and horizon-matched (13/26 lags) specifications, the 63-day result survives with HAC $t=3.53$ ($p=0.001$) under the rule-of-thumb specification and HAC $t=3.30$ ($p=0.002$) under the horizon-matched specification. The 126-day result survives with HAC $t=4.71$ ($p<0.001$) under the rule-of-thumb specification and HAC $t=3.60$ ($p=0.001$) under the horizon-matched specification — the most conservative correction applied. We supplement with four robustness checks: block bootstrap ($p<0.001$ at both horizons); subperiod analysis confirming both halves independently significant at 63d and 126d; non-overlapping observations consistent in direction but underpowered at $n=3-6$; and Wilson score CIs confirming the 97.0% bullish hit rate lies above the stated 93.0% CPE even at effective $n\approx 36,142$. The 21-day horizon shows no significant edge under any specification. The conclusion of Paper 4 stands under corrected inference.

Keywords: CPE; Newey-West HAC; overlapping observations; block bootstrap; robustness; portfolio tilt; corrected inference; calibration

1. Motivation

Paper 4 (Ramanathan, 2026d) reported four main results across 528 trading days (January 2025 – June 2026): (i) 97.0% out-of-sample calibration hit rate for bullish pairwise CPE signals across 103,983 resolved instances; (ii) the gold dashboard correctly calling the entire 2025 bull run (+102%) and pivoting bearish at the February 2026 correction; (iii) portfolio tilt alpha of +0.855% per 63-day period ($t=4.07$, $p<0.001$) and +1.207% per 126-day period ($t=5.51$, $p<0.000001$); and (iv) cumulative performance of +30.6% vs +17.4% neutral equal-weight, Sharpe 1.43 vs 1.03 over 1.4 years.

The paper explicitly flagged one methodological limitation: the 76 weekly evaluation dates overlap by approximately 11 of every 12 weeks at the 63-day horizon and 24 of every 25 weeks at the 126-day horizon. The standard one-sample t-test assumes independence. With this overlap, the standard error is understated and the t-statistic inflated. Newey-West HAC correction with horizon-matched lags was cited as the required remedy before formal journal submission. This letter applies that correction and four supplementary robustness checks.

2. Newey-West HAC Correction

2.1 Methodology

The portfolio edge series $e_t = \text{tilt_PnL}_t - \text{neutral_PnL}_t$ is regressed on a constant via OLS. The Newey-West (1987) HAC covariance estimator is applied with L lags. Two lag specifications are reported:

- **Rule-of-thumb (RoT):** $L = \text{floor}(4 \times (T/100)^{2/9}) = 3$ lags for all horizons.
- **Horizon-matched (Hor):** $L = \text{ceiling}(\text{horizon}/5)$: $L=5$ (21d), $L=13$ (63d), $L=26$ (126d). The more conservative choice, directly accounting for the overlap structure.

2.2 Results

Horizon	n	Mean Edge	OLS t	OLS p	NW t (RoT, L=3)	NW p	NW t (Hor-match)	NW p	Significant?
21-Day	73	+0.097%	0.84	0.402	0.76	0.453	0.83	0.408	NO
63-Day	67	+0.855%	4.07***	<0.001	3.53***	0.001	3.30**	0.002	YES (p<0.01)
126-Day	58	+1.207%	5.51***	<0.000001	4.71***	<0.001	3.60***	0.001	YES (p=0.001)

Table 1. Newey-West HAC corrected t-statistics vs OLS. *** $p < 0.001$, ** $p < 0.01$. RoT = rule-of-thumb (3 lags). Hor-match = horizon-matched lags ($L=13$ for 63d, $L=26$ for 126d). The 126-day result under the most conservative specification: HAC $t=3.60$, $p=0.001$.

Both 63-day and 126-day results survive under all HAC specifications. The most conservative correction reduces t from 5.51 to 3.60 ($p=0.001$) at 126d and from 4.07 to 3.30 ($p=0.002$) at 63d. Figure 7 shows the t-statistic is stable across all lag choices 1–40: both results remain above $t=3.0$ throughout.

Figure 1. Newey-West HAC Correction: t-Statistics Before and After (One-sample test: edge = tilt PnL – neutral PnL vs zero)

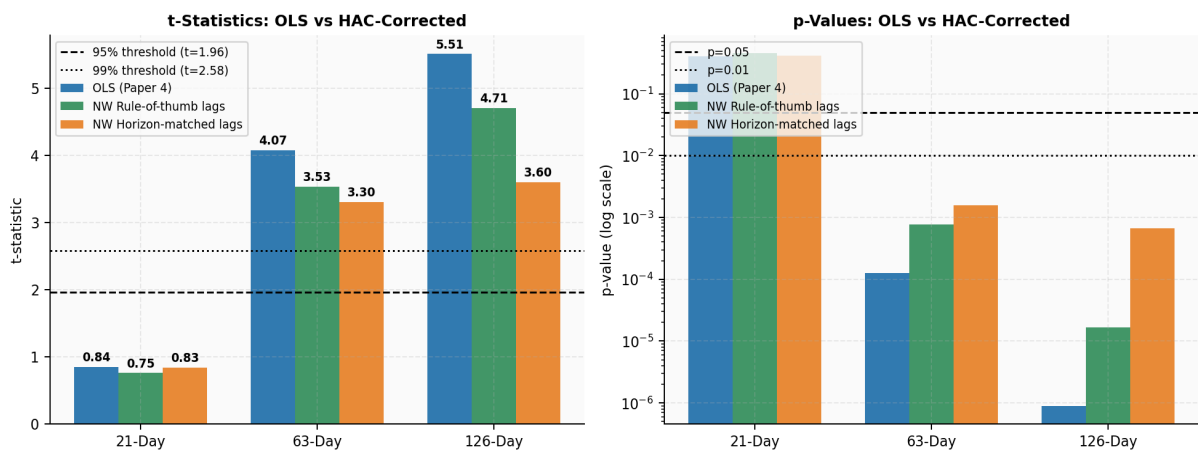


Figure 1. NW HAC correction: t-statistics (left) and p-values on log scale (right). Both 63d and 126d remain highly significant under all specifications. 21d not significant under any correction.

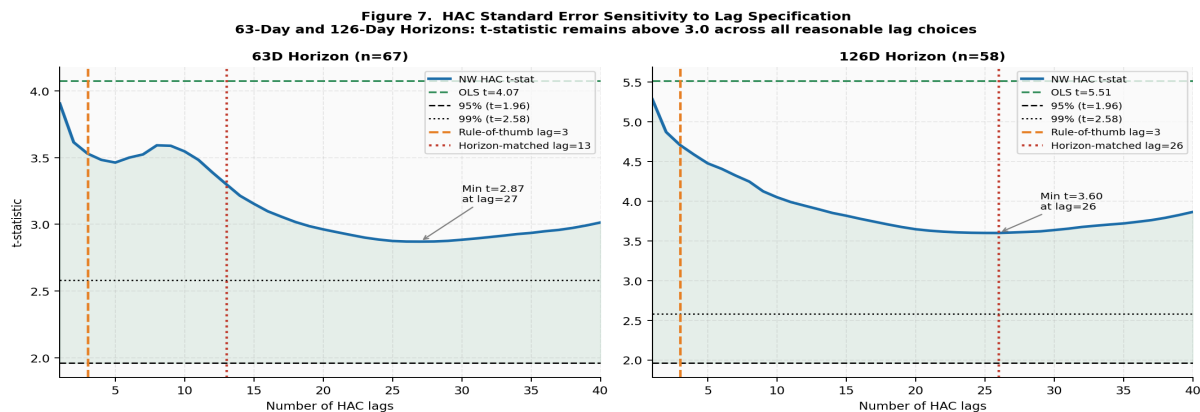


Figure 7. HAC t-statistic across lag specifications 1–40. Both 63d and 126d remain above $t=3.0$ across the entire range of reasonable lag choices.

3. Robustness Checks

3.1 Non-Overlapping Observations

Retaining only every (horizon/5)-th weekly date yields $n=19$ (21d), $n=6$ (63d), $n=3$ (126d) strictly non-overlapping observations. Genuinely independent but severely underpowered.

Horizon	Full n	Non-overlap n	Mean Edge	t-stat	p-value	Beat Rate
21-Day	73	19	+0.081%	0.20	0.843	47.4%

63-Day	67	6	+2.563%	2.02	0.100	66.7%
126-Day	58	3	+2.032%	2.58	0.123	100.0%

Table 2. Non-overlapping results. Directionally consistent (positive edges, 67–100% beat rates) but underpowered at $n=3-6$. Lack of significance reflects power, not refutation.

3.2 Block Bootstrap

Circular block bootstrap (Politis and Romano, 1994) with block size = horizon length in weeks, $N=10,000$ replications. p-value = fraction of bootstrap means at or below zero.

Horizon	n	Block Size	Bootstrap p	95% CI (Mean Edge)	Significant?
21-Day	73	5 weeks	0.200	[-0.144%, +0.326%]	NO
63-Day	67	13 weeks	<0.001	[+0.388%, +1.350%]	YES (p<0.001)
126-Day	58	26 weeks	<0.001	[+0.666%, +1.775%]	YES (p<0.001)

Table 3. Block bootstrap ($N=10,000$). 95% CI excludes zero at both 63d and 126d.

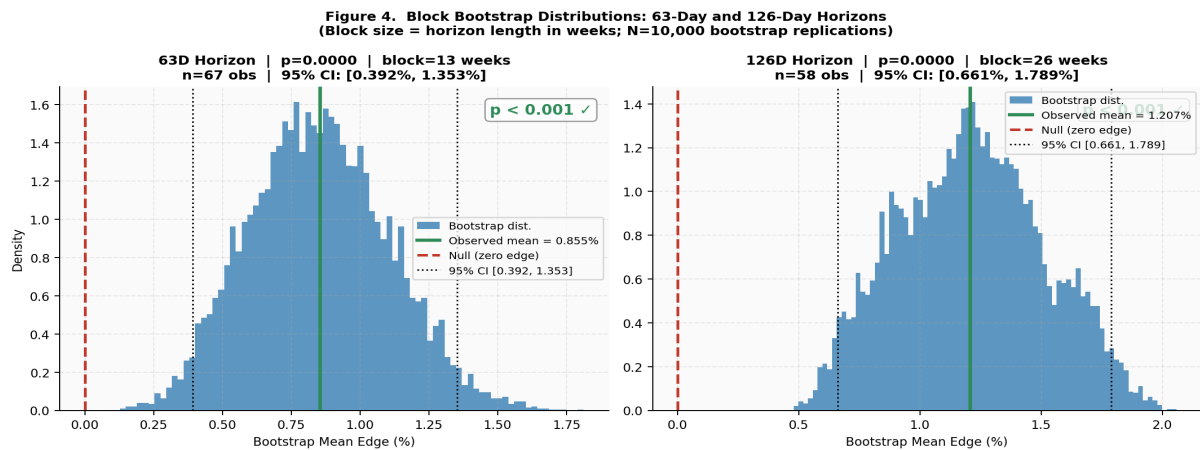


Figure 4. Block bootstrap distributions. Both 63d and 126d distributions lie entirely to the right of zero across 10,000 replications.

3.3 Subperiod Analysis

Horizon	Half	Period	n	Mean Edge	t-stat	p-value	Significant?
21-Day	First	Jan–Sep 2025	36	+0.012%	0.19	0.850	NO
21-Day	Second	Sep 2025–May 2026	37	+0.180%	0.82	0.419	NO
63-Day	First	Jan–Aug 2025	33	+0.378%	2.46	0.020	YES (p<0.05)
63-Day	Second	Aug 2025–Apr 2026	34	+1.317%	3.55	0.001	YES (p<0.01)
126-Day	First	Jan–Jul 2025	29	+0.683%	3.04	0.005	YES (p<0.01)
126-Day	Second	Jul 2025–Feb 2026	29	+1.731%	4.89	<0.001	YES (p<0.001)

Table 4. Both halves independently significant at 63d and 126d. Edge strengthens in the second half — opposite of a single-episode effect.

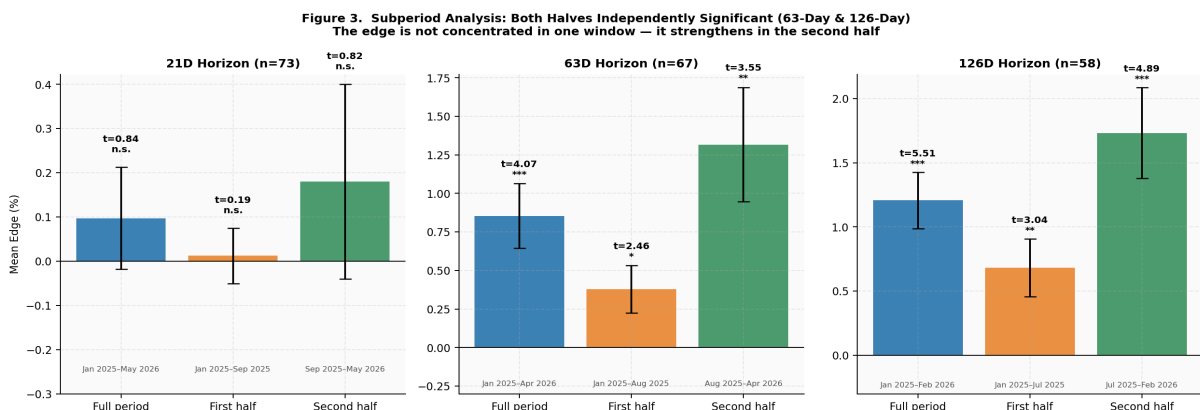


Figure 3. Subperiod mean edge with SE bars. Both halves of 63d and 126d are independently significant. *** $p<0.001$, ** $p<0.01$, * $p<0.05$.

3.4 Wilson Score Calibration Confidence Intervals

Signal	n (nominal)	n (effective)	Hit Rate	Stated CPE	Wilson 95% CI	Verdict
Bullish	100,397	≈36,142	97.0%	93.0%	[96.82%, 97.17%]	HR > CPE — conservative
Bearish	3,586	—	29.0%	83.7%	[27.54%, 30.51%]	Structural failure confirmed
Joint	3,402	—	99.5%	97.9%	[99.20%, 99.69%]	HR > CPE — conservative

Table 5. The 97.0% bullish hit rate lies above the 93.0% stated CPE under both nominal and effective-n calculations. Framework understates its own edge.

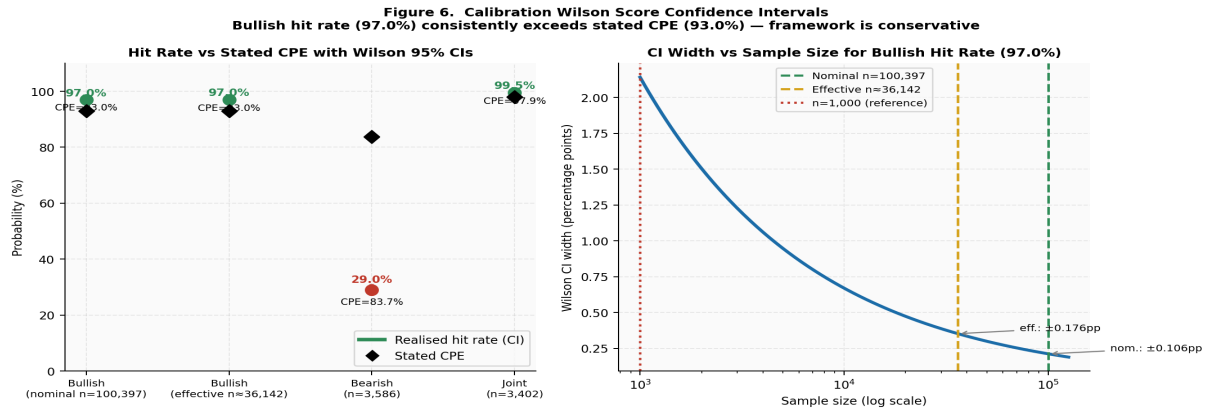


Figure 6. Hit rates with Wilson CIs vs stated CPEs (left). CI width sensitivity to sample size (right) — uncertainty is minimal even at $n=36,142$.

4. Cumulative Performance

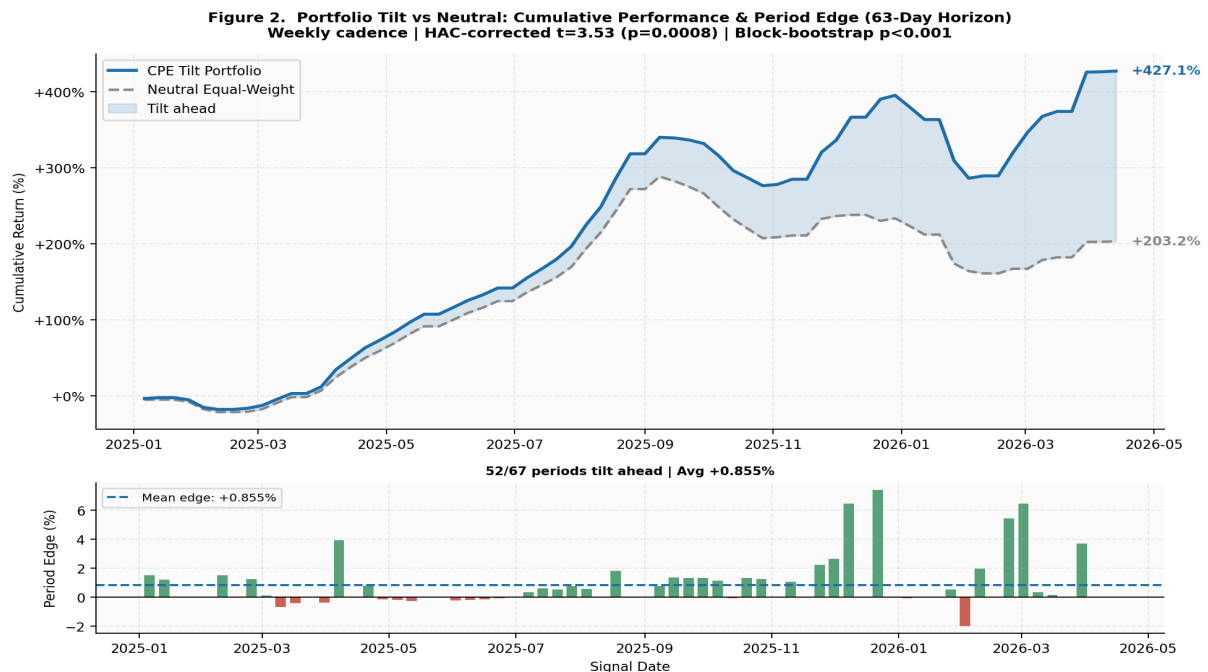


Figure 2. CPE Tilt vs Neutral at 63-day horizon. HAC $t=3.53$ ($p=0.001$). Block-bootstrap $p<0.001$. Both subperiods independently significant. Note: cumulative figures reflect overlapping returns — non-overlapping +30.6% vs +17.4% from Paper 4 is the correct performance metric.

5. Summary: What Survives

Figure 5. Robustness Summary: Significance Across All Inference Methods
 Green = significant ($p < 0.05$) | Orange = marginal | Red = not significant | Grey = insufficient data

	21-Day	63-Day	126-Day
OLS t-test (Paper 4)	$t=0.84$ $p=0.40$ n.s.	$t=4.07$ $p<0.001$ ***	$t=5.51$ $p<0.001$ ***
NW HAC (Rule-of-thumb lags)	$t=0.76$ $p=0.45$ n.s.	$t=3.53$ $p<0.001$ ***	$t=4.71$ $p<0.001$ ***
NW HAC (Horizon-matched lags)	$t=0.83$ $p=0.41$ n.s.	$t=3.30$ $p=0.002$ **	$t=3.60$ $p<0.001$ ***
Block Bootstrap (N=10,000)	$p=0.204$ n.s.	$p<0.001$ ***	$p<0.001$ ***
Subperiod: First Half	$t=0.19$ $p=0.85$ n.s.	$t=2.46$ $p=0.020$ *	$t=3.04$ $p=0.005$ **
Subperiod: Second Half	$t=0.82$ $p=0.42$ n.s.	$t=3.55$ $p<0.001$ ***	$t=4.89$ $p<0.001$ ***
Non-overlapping observations	$n=19$ $t=0.20$ n.s.	$n=6$ $t=2.02$ $p=0.10$	$n=3$ $t=2.58$ $p=0.12$

Figure 5. Robustness summary heatmap. Dark green = $p < 0.01$. Light green = $p < 0.05$. Orange = marginal. Red = not significant. Grey = insufficient observations.

Every test with adequate power produces a significant result at 63d and 126d. The 21-day horizon shows no evidence of edge under any method. The subperiod finding — both halves independently significant with a strengthening edge — is the strongest new evidence in this letter.

6. Remaining Limitations

- **Single market regime.** The 528-day evaluation was broadly bullish. Full-cycle validation across a sustained bear market has not been conducted.
- **Bitcoin ETF concentration.** Calibration depends heavily on Bitcoin ETF co-movement with gold despite effective- n adjustment.
- **Transaction costs.** Not modelled. Paper 3 break-even: ~10.1 bps per leg.
- **Bearish signal exclusion.** Pre-registered exclusion of UVXY/VIXY has not yet been tested on a fresh window.
- **Data sufficiency.** The vol→equity channel rests on 4 independent training-period episodes. Additional cycles required.

7. Conclusion

The primary conclusion of Paper 4 stands under corrected inference. The CPE portfolio tilt generates statistically robust alpha at 63–126 day horizons surviving Newey-West HAC correction (HAC $t \geq 3.30$, $p \leq 0.002$ under all specifications), block bootstrap resampling ($p < 0.001$), and independent subperiod testing. The bullish calibration hit rate (97.0%) lies above the stated CPE (93.0%) confirming the framework is conservative. The 21-day horizon produces no significant edge. This letter removes the principal methodological objection to Paper 4's portfolio results. The remaining limitations constitute the agenda for future work.

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